

Probability distributions

For discrete X : $P(X=x_i) = P_i$ and
 $\sum P_i = 1$

For continuous X : density $\rightarrow f(x)$
then $\int f(x) dx = 1$

Ex:-

1) $X =$ no of heads in 3 coin tosses

Solⁿ:- Sample space

$S = \{HHH, HHT, HTH, HTT, TTT, TTH, THT, TTH\}$

No of heads may be 0, 1, 2, 3

$\therefore$ Range is $\{0, 1, 2, 3\}$

$$P(X=0) = \frac{1}{8} = P_1$$

(Probability of getting zero heads)

$$P(X=1) = \frac{3}{8} = P_2$$

$$P(X=2) = \frac{3}{8} = P_3$$

$$P(X=3) = \frac{1}{8} = P_4$$

$$P_1 + P_2 + P_3 + P_4 = 1$$

Ex 2:- $f(x) = 2x$ for $0 < x < 1$. Verify its a probability distribution function and find $P(X < 1/2)$

Soln:-

To check it is pdf.

(i) $f(x) = 2x$ if $0 < x < 1$

$\therefore$ it is ≥ 0

if $x \leq 0$ $f(x) = 0$

if $x \geq 1$ $f(x) = 0$

$\therefore f(x) \geq 0 \quad \forall x$

(ii)
$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 f(x) dx = \int_0^1 2x dx$$
$$= x^2 \Big|_0^1 = 1$$

$\Rightarrow$ Area under the curve = 1.

$\therefore f$ is pdf.

To compute $P(X < 1/2)$

$$P(X < 1/2) = \int_{-\infty}^{1/2} f(x) dx = \int_0^{1/2} f(x) dx$$

(since for $x < 0$, $f(x) = 0$)

$$\Rightarrow P(X < 1/2) = \int_0^{1/2} 2x \, dx = x^2 \Big|_0^{1/2} = \frac{1}{4}$$

Expectation

$$E[X] = \begin{cases} \sum x_i p_i, & \text{if } X \text{ is discrete} \\ \int x f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Moment generating function

$$M_X(t) = E[e^{tx}] = \begin{cases} \sum e^{tx_i} p_i & \text{discrete} \\ \int e^{tx} f(x) dx & \text{continuous} \end{cases}$$

Ex:-

$$1) \quad P(X=0) = P(X=1) = 1/2$$

Here $x_1 = 0$, $x_2 = 1$

$$p_1 = P(X=0) = 1/2 \quad p_2 = P(X=1) = 1/2$$

$$E(X) = x_1 p_1 + x_2 p_2 = 0 \times 1/2 + 1 \times 1/2 = 1/2$$

$$\begin{aligned} M_X(t) &= e^{tx_1} p_1 + e^{tx_2} p_2 \\ &= e^{t(0)} \frac{1}{2} + e^{t(1)} \frac{1}{2} \\ &= \frac{1}{2} + e^t \frac{1}{2} = \frac{1+e^t}{2} \end{aligned}$$

x with pdf $f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Soln:-

$$E[x] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x(2x) dx$$

$$= 2 \int_0^1 x^2 dx = 2 \left(\frac{x^3}{3} \right) \Big|_0^1 = 2 \left(\frac{1}{3} \right) = \frac{2}{3}$$

$$E[x] = \frac{2}{3}$$

MGF

$$M_x(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^1 e^{tx} (2x) dx$$

$$\Rightarrow M_x(t) = 2 \int_0^1 x e^{tx} dx$$

$$= 2 \left\{ x \int_0^1 e^{tx} dx - \int_0^1 \frac{d}{dx} (x) \int e^{tx} dx \right\}$$

$$= 2 \left\{ \left(x \frac{e^{tx}}{t} \right) \Big|_0^1 - \int_0^1 \frac{e^{tx}}{t} dx \right\}$$

$$= 2 \left\{ \left(\frac{e^t}{t} - 0 \right) - \left(\frac{e^{tx}}{t^2} \right) \Big|_0^1 \right\}$$

$$= 2 \left\{ \frac{e^t}{t} - \left(\frac{e^t}{t^2} - \frac{1}{t^2} \right) \right\}$$

$$\Rightarrow M_x(t) = \frac{2(e^t(t-1)+1)}{t^2}$$

